# A boundary only by the clock: the black-hole horizon between natural and exit, the membrane paradigm as the minimal semigroup, and the price of exterior unitarity

Jonathan Torkelson
Embeddetech, Inc., Tulsa, Oklahoma, USA
jtorkelson@embeddetech.com

July 19, 2026

**Abstract**

**Draft — not for circulation.**

The first paper of this series priced incomplete dynamics in a single currency: for a half-line Schrödinger or diffusion problem, one classical time-of-flight integral simultaneously decides classical completeness, the Weyl limit-point/limit-circle alternative, and the Feller boundary type; conservative dynamics beyond the limit-point regime is a *choice* of self-adjoint extension (boundary data); and the choice-free, regulator-independent object is the minimal sub-stochastic semigroup. This paper runs that dictionary at a black-hole horizon, where it acquires an axis the non-relativistic theory did not have: the classification is *clock-dependent*. Three results, for a scalar field and its radial diffusion analogue on fixed Schwarzschild (with Rindler as warm-up). (i) *A clock-dependence theorem.* In Killing time the horizon is Weyl limit point and a Feller *natural* boundary: infinite time-of-flight, essentially self-adjoint generator, conservative minimal semigroup, no room for boundary data. In horizon-adapted clocks (Painlevé–Gullstrand, infalling proper, hyperboloidal) the time-of-flight is finite and the horizon generically becomes an *exit* boundary (unless the noise can climb an e-folding of the lapse; see below): the exterior generator is only dissipative (non-normal), its minimal semigroup is sub-stochastic, and the lost mass is exactly the flux through the horizon. The two pictures are related by a singular time change whose rate is set by the lapse, and Feller boundary type is not invariant under singular time changes. The stochastic half is proved here—with an exact identity equating Feller's attainability integral to the clock's ingoing time-of-flight—while the quantum half is assembled from known limit-point facts and the hyperboloidal-slicing program, cited rather than re-derived. A byproduct of the proof: the finer exit-versus-regular alternative is decided by the *outgoing* integral, and the boundary-data (limit-circle-like) regime opens only when the noise can climb an e-folding of the lapse—the stochastic twin of the near-horizon conformal-quantum-mechanics regime of the literature, and a feature of the isotropic clock idealization that the body argues does not survive a genuine (anisotropic) graph clock. (ii) *The membrane paradigm as the minimal process.* Stretched-horizon completions—brick-wall Dirichlet walls, near-horizon $U(1)$ extension families, antipodal S-matrix identifications—are conservative dilations carrying boundary data at a regularized wall; they are mutually equivalent for exterior scattering and thermality—rigorously for the $U(1)$ extension family, conjecturally in the multichannel (antipodal) case—but inequivalent for entropy counting, which is exactly the brick wall's

notorious cutoff dependence, read as extension dependence. The wall-independent object is the minimal sub-stochastic semigroup, and wall-removal universality (for local wall conditions) is the exit property (strictly, its interior $\kappa pD < v_0$). (iii) *A dictionary entry for the information paradox*—not a resolution. "Does the horizon need extra data?" has a clock-indexed answer: no in Killing time; in infalling time only if one insists on exterior conservativity, in which case the data parameterize inequivalent unitary completions. Firewall-type tensions, insofar as they demand a single clock-independent verdict on what the horizon carries, ask a question that—at the level of a test field's boundary type—has no clock-independent answer: provably so for the radial diffusion analogue, and at the cited strength of the hyperboloidal program for the field itself.

**Note on provenance.** This manuscript is an experiment in AI-conducted research as much as a physics paper. The project began when the author asked the large language model Claude Fable 5 (Anthropic): *"Is it not possible that the antilinear PT/time-reversal symmetry element could maybe predict indivisible non-Markovian processes?"* Everything that followed was performed by the model: the research program, the theorems and proofs, the numerical experiments, the literature search, and this text. The model was repeatedly prompted to attack its own results adversarially and to search for prior work. The author, a professional engineer rather than a physicist, directed and reviewed the sessions but makes no claim to comprehension of the findings, and is circulating the manuscript for independent review by professional physicists. Accordingly, nothing here should be taken on trust: every result is stated with the proof it actually has—here entirely closed-form derivations and cited results, with no numerical experiments; the originating conversation, project briefing, and full research notebook are preserved in the program's hub repository, `https://github.com/embeddetech/ghosts-program`.

# Contents


**1 Introduction** **4**
- 1.1 The question . . . 4
- 1.2 Results . . . 5
- 1.3 What is not claimed, and whose results are load-bearing . . . 7

**2 The explosion dictionary on a half-line, recalled** **7**

**3 Warm-up: the Rindler wedge** **8**
- 3.1 Geometry and clocks . . . 8
- 3.2 The Killing pair and its operator . . . 9
- 3.3 The adapted pair and the inverse-square foreshadowing . . . 9
- 3.4 The stochastic columns and the global dilation . . . 9

**4 Schwarzschild: radial reduction, weights, and clocks** **10**
- 4.1 Reduction with the weight kept honest . . . 10
- 4.2 Clocks and their time-of-flight integrals . . . 10

**5 The clock-dependence theorem** **11**
- 5.1 The stochastic model . . . 11
- 5.2 The quantum column: proved and imported . . . 14
- 5.3 Arbitrating limit point versus limit circle at the horizon . . . 15
- 5.4 States versus domains; which "clock" . . . 16

**6 The membrane paradigm as the minimal process** **16**
- 6.1 The membrane paradigm, restated . . . 16
- 6.2 Walls, completions, and the wall-removal limit . . . 17
- 6.3 The dictionary of completions . . . 18
- 6.4 Scattering equivalence, entropy inequivalence . . . 18

**7 The information paradox as a dictionary entry** **19**

**8 Outlook** **20**


# 1 Introduction

## 1.1 The question

Does a black-hole horizon carry boundary data? The question splits the literature into camps that rarely cite one another's mathematics. On one side, the folklore of the tortoise coordinate: the radial wave equation outside a Schwarzschild black hole, written in Killing time, is a Schrödinger problem on the whole line whose potential dies exponentially toward the horizon; the operator is essentially self-adjoint; nothing may be prescribed at the horizon, and the familiar "purely ingoing" condition of quasinormal-mode theory is an asymptotic regularity selection, not a boundary condition in the extension-theoretic sense. On another side, the near-horizon conformal-quantum-mechanics literature finds, in a proper-distance-like variable, an inverse-square operator in its limit-circle regime and a $U(1)$ family of self-adjoint extensions whose choice feeds the entropy

[19, 20, 18]. On a third, the membrane paradigm [23, 24, 25] treats the horizon as a dissipative fluid membrane—manifestly *not* conservative dynamics—while 't Hooft's brick wall [27] and antipodal identification [28] restore unitarity by installing data at or near the horizon. Each camp is internally consistent. Their mutual tension—no data possible, a family of data required, dissipation instead of data, data restoring unitarity—is the subject of this paper.

The claim defended here is that all four camps are right, because they are answering four different well-posed questions, and that a single piece of arithmetic separates the questions: the classical time-of-flight integral to the horizon, *evaluated with a specified clock*. The framework is the explosion dictionary of the first paper of this series [1], which established, for half-line Schrödinger and diffusion problems, a three-column correspondence: (i) finite classical time-of-flight to an endpoint signals loss of essential self-adjointness (Weyl limit circle); (ii) conservative dynamics then requires a choice of self-adjoint extension—boundary data, a $U(1)$ phase or, in multichannel form, a unitary S-matrix at the endpoint; (iii) the choice-free, regulator-independent object is the minimal Feller semigroup, which is sub-stochastic (mass leaks through the endpoint); and (iv) one time-of-flight integral decides classical completeness, the Weyl alternative, and the Feller boundary type (regular/exit/entrance/natural) simultaneously, for the potential classes treated there.

At a horizon this dictionary meets a genuinely new phenomenon. The time-of-flight to the horizon is *infinite* in Killing time and *finite* in the proper time of an infalling observer—the oldest fact in black-hole physics. In the dictionary's terms this is not a curiosity but a theorem generator: the boundary classification of the horizon is *clock-dependent*, the two classifications are related by a singular time change (the rate degenerating like a power of the metric function $f$), and Feller boundary type is not invariant under singular time changes. The horizon is the starkest example the dictionary allows: a single endpoint that is a natural boundary for one clock and an exit boundary for another.

### 1.2 Results

Throughout, the setting is a test scalar field, and a radial transport diffusion modeled on it, on the *fixed* exterior Schwarzschild geometry (Section 4), with the Rindler wedge as warm-up (Section 3). Backreaction, evaporation, and any holographic interpretation are outside the scope; see the scope statement in Section 1.3.

**S1 (clock-dependence theorem, Section 5).** For the exterior radial problem:

- In *Killing time* the horizon is Weyl limit point and a Feller *natural* boundary: the time-of-flight diverges (logarithmically, rate set by the surface gravity $\kappa$); the spatial generator is essentially self-adjoint (Proposition 5.9); the minimal diffusion semigroup is conservative; no boundary condition can be imposed (Theorem 5.3(a)).

- In *horizon-adapted clocks* (Painlevé–Gullstrand, infalling proper, hyperboloidal) the time-of-flight is finite and the horizon becomes an *exit* boundary in the generic regime $\kappa p D \leq v_0$ (the complementary regime is regular, Theorem 5.3(c)): the process reaches it almost surely in finite clock time, cannot return, the minimal semigroup is strictly sub-stochastic with mass deficit equal to the accumulated probability flux through the horizon, and every conservative completion requires adjoining horizon degrees of freedom (Theorem 5.3(b,c), Corollary 5.4).

- The two descriptions are related by a singular time change with rate degenerating like a power of the lapse function $f$ (the metric function $g_{tt} = -f$; the ADM lapse proper is $\sqrt{f}$, the Rindler $\rho$ of the warm-up, so "a power of $f$" is that power of the squared lapse); Feller type is not invariant under such time changes. The decider is exact, not merely asymptotic: Feller's attainability integral for the time-changed process *equals* the adapted clock's ingoing time-of-flight integral, for every admissible rate function (equation (15)).

- A finer statement falls out of the proof. The exit-versus-regular alternative at the horizon in adapted time is decided by the *outgoing* integral: the boundary is exit iff $\kappa p D \leq v_0$, where $D$ is the noise strength, $v_0$ the ingoing transport speed, and $p$ the lapse power of the clock rate. Equivalently, boundary data become possible in adapted time exactly when the noise's upstream penetration depth $D/v_0$ exceeds the lapse e-folding length $1/\kappa p$. Macroscopic horizons sit deep in the exit regime; the regular regime is the stochastic twin of the near-horizon limit-circle (conformal quantum mechanics) regime of [19, 20, 18] (Remark 5.5)—though it is a feature of the isotropic clock idealization that the body argues (at sketch strength) does not survive a genuine graph clock (Remark 5.7(iii)).

The stochastic half of S1 is proved in full. The quantum half is deliberately assembled from the literature: essential self-adjointness in the Killing pair is proved here because the proof is three lines, while the dissipativity of the horizon-adapted *quantum* generator is the hyperboloidal-slicing program of Warnick [15] and Jaramillo–Panosso Macedo–Al Sheikh [17], imported as Proposition 5.10 and not re-derived. The honest formulation of "changing the clock" (the Painlevé–Gullstrand time function is generated by the *same* Killing field) is discussed in Section 5.4: what changes is the parameterization of the transported process, respectively the (Hilbert space, slicing) pair, and it is the diffusion version that is theorem-grade today.

**S2 (membrane paradigm = minimal process, Section 6).** In horizon-adapted time the exterior problem has a canonical choice-free description: the minimal sub-stochastic semigroup. We identify this object with the membrane paradigm: dissipative, positivity-preserving, contractive, with the membrane's absorption law as the mass-flux identity, and regulator-independent in a precise sense—it is the monotone limit of absorbing-wall (stochastic Dirichlet) regularizations, and in the exit regime (strictly, $\kappa p D < v_0$) *every local* wall condition (Dirichlet, Neumann, elastic, sticky) converges to it as the wall is removed (Proposition 6.1); non-local re-injection (jump) walls, which return lost mass to the interior, are the exception. Unitary or conservative completions —'t Hooft's brick wall, the Balachandran–de Queiroz–Saa $U(1)$ extension family, the antipodal S-matrix, and (in the Rindler warm-up) the global Minkowski dynamics itself—are conservative dilations carrying boundary data at a stretched horizon; on exterior scattering and thermality they agree with each other—rigorously for the $U(1)$ extension family (the equivalence lemma of [18]), conjecturally in the multichannel/antipodal case—and with the minimal object itself only in the scalar diffusion analogue (Proposition 6.1(ii), one domination step at sketch strength; the re-injection walls excepted)—and disagree on entropy counting, which is exactly the brick wall's cutoff dependence read as extension dependence. No entropy theorem is claimed; the entropy statements are dictionary readings of cited results.

**S3 (information-paradox dictionary entry, Section 7).** A carefully fenced discussion, structural rather than resolving: "does the horizon need extra data?" is clock-indexed. In Killing time, no—exterior unitarity is the unique conservative dynamics and involves no choice. In

infalling time, data are generically optional: refusing them yields the minimal sub-stochastic description (information loss *for the exterior algebra*); insisting on exterior conservativity forces a choice of dilation data at the horizon, and inequivalent choices exist. Firewall-type arguments [29], *insofar as* they presuppose a single clock-independent classification of what the horizon carries, presuppose something that—by S1, at the level of a test field's boundary type—does not exist; whether they in fact require it is a quantum-gravity question this dictionary leaves open.

### 1.3 What is not claimed, and whose results are load-bearing

Three neighboring results are used and must not be confused with the contribution of this paper.

First, *conservative dynamics = choice of positive self-adjoint extension* in static spacetimes is Wald's 1980 prescription [3] together with the Ishibashi–Wald classification theorems [4, 5]: any conservative, locally determined, energy-conserving dynamics in a static region corresponds to some positive self-adjoint extension of the spatial operator. They applied this to naked singularities and to the AdS timelike boundary; the horizon instance in Killing time is the trivial case (the extension is unique). Our quantum column stands entirely on their template.

Second, *dissipative horizon ⇒ non-self-adjoint generator* is the hyperboloidal/pseudospectrum program [15, 16, 17]: quasinormal modes as genuine eigenvalues of a non-self-adjoint generator on horizon-crossing slicings, with non-normality tied to losses through the boundary. We import this as the quantum half of the adapted-clock column.

Third, *the near-horizon $U(1)$ extension family and the brick wall* belong to Gupta–Sen [19], Camblong–Ordóñez [20, 21], and Balachandran–de Queiroz–Saa [18], the last explicitly framing the extension family as a generalization of the brick wall and proving the scattering/thermality equivalence of extensions.

What this paper adds is the column none of these literatures has: the *probabilistic* one—the Feller boundary classification of the horizon, which appears to be genuinely absent from the literature— together with the single time-of-flight integral as the cross-column decider and the clock-dependence theorem that the three columns jointly enforce. The scope guard, stated once and observed throughout: a test scalar field and its radial diffusion analogue on fixed Schwarzschild (plus Rindler); no evaporation, no backreaction, no holography.

## 2 The explosion dictionary on a half-line, recalled

We recall from [1] the minimum needed here; proofs and the full dictionary are in that paper and the classical references [10, 11, 12, 9].

**Weyl alternative.** For a Sturm–Liouville operator $A = -\mathrm{d}^2/\mathrm{d}x^2 + V$ on an interval, each endpoint is either *limit point* (LP)—no boundary condition possible or needed—or *limit circle* (LC)—a one-parameter $U(1)$ family of boundary conditions, each yielding a distinct self-adjoint extension and hence a distinct conservative quantum dynamics. If both endpoints are LP, $A$ is essentially self-adjoint on $C_c^\infty$: the conservative dynamics is unique. A sufficient LP criterion used repeatedly below: at an endpoint at infinite coordinate distance, $V$ bounded below suffices [9, Thm. X.8 and Appendix to X.1].

**Feller classification.** For a one-dimensional diffusion with scale function $s$ and speed measure $m$ on $(l, r)$, fix an interior reference point $x_0$ and define, for the left endpoint $l$,

$$\Sigma(l) = \int_l^{x_0} \big[\mathcal{M}(x_0) - \mathcal{M}(\xi)\big]\, \mathrm{d}s(\xi), \qquad N(l) = \int_l^{x_0} \big[s(x_0) - s(\eta)\big]\, \mathrm{d}\mathcal{M}(\eta), \tag{1}$$

with $\mathcal{M}$ any primitive of $m$. Then $l$ is *regular* if $\Sigma < \infty$ and $N < \infty$ (attainable and enterable: boundary conditions from Feller's lateral-condition menu—absorbing, reflecting, elastic, sticky—parameterize the possible dynamics); *exit* if $\Sigma < \infty$, $N = \infty$ (attainable, not enterable: the process can be lost through $l$ but no lateral condition on the open interval restores conservativity); *entrance* if $\Sigma = \infty$, $N < \infty$ (enterable, not attainable); *natural* if both diverge (the boundary is dynamically infinitely far: unattainable, unenterable, no data possible) [10, 11, 12].

**Minimal semigroup.** On the open interval, killing the process at the endpoints defines the *minimal* semigroup $P_t^{\min}$: positivity-preserving, contractive, sub-stochastic. It is the smallest positive semigroup compatible with the local generator, is independent of any regulator, and every conservative dynamics dominates it. When an endpoint is exit, $1 - P_t^{\min}\mathbf{1}(x) = \mathbb{P}_x[\text{lost through the endpoint by } t] > 0$: the deficit is a boundary flux, and conservativity can be restored only by enlarging the state space (adjoining the endpoint as an absorbing cemetery, or reinjecting). In [1] this object is the regulator-free description of an exploding quantum/stochastic system, and self-adjoint extensions are its conservative completions.

**Time-of-flight decider.** For the half-line problems of [1], the single integral $T = \int^x \mathrm{d}x'/v(x')$ (classical time-of-flight to the endpoint at local speed $v$) decides all columns at once: $T = \infty$ gives LP/natural (no data), $T < \infty$ gives LC/accessible (data or loss). (The equivalence holds for the potential classes of [1]; a *repulsive* inverse-square tail $+c/x^2$ with $0 < c < \frac{3}{4}$ is the textbook exception—the classical barrier makes the endpoint unattainable, $T = \infty$, yet it is limit circle—so $T = \infty \Rightarrow$ LP is not universal. The near-horizon operator of Section 3.3 is the opposite, decider-consistent case: *attractive* coupling, finite $T$, limit circle.) The new feature at horizons is that $v$—hence $T$, hence everything—depends on the clock.

# 3 Warm-up: the Rindler wedge

## 3.1 Geometry and clocks

The right Rindler wedge of Minkowski space, $X > |T|$, carries coordinates $(\eta, \rho)$ with $T = \rho \sinh \eta$, $X = \rho \cosh \eta$ and metric

$$\mathrm{d}s^2 = -\rho^2\, \mathrm{d}\eta^2 + \mathrm{d}\rho^2 + \mathrm{d}y_\perp^2. \tag{2}$$

The boost field $\partial_\eta$ is the wedge's Killing time; the horizon is $\rho = 0$; the lapse is $\rho$ (unit surface gravity in these conventions). Two clocks present themselves: boost time $\eta$ (the wedge-static observer's clock, rescaled) and inertial Minkowski time $T$ (the clock of an observer who falls through the horizon).

The classical time-of-flight already splits. A radial null ray has $\mathrm{d}\rho/\mathrm{d}\eta = \pm\rho$, so the boost-time flight to the horizon from $\rho_0$ is $\int_0^{\rho_0} \mathrm{d}\rho/\rho = \infty$ (logarithmic divergence—the tortoise coordinate $x = \ln \rho$ is this divergence made into a coordinate). The same ray reaches $\rho = 0$ at finite inertial time $T$. Along the ingoing ray $\rho = \rho_0 \mathrm{e}^{-\eta}$, so the inertial clock rate relative to boost time is $\mathrm{d}T/\mathrm{d}\eta = \rho\, \mathrm{e}^{-\eta} = \rho^2/\rho_0 \sim \rho^2$: quadratic in the boost lapse $\rho$ (equivalently linear in $f = \rho^2$, matching the $p = 1$ of the infalling clock).

### 3.2 The Killing pair and its operator

A massive scalar with transverse momentum $k_\perp$, in boost time $\eta$, separates as $\phi = \mathrm{e}^{-\mathrm{i}\omega\eta}\psi(\rho)$ with

$$A_R\,\psi = -\rho\,\partial_\rho(\rho\,\partial_\rho\psi) + \mu^2\rho^2\psi = \omega^2\psi, \qquad \mu^2 = m_\phi^2 + k_\perp^2, \tag{3}$$

where $A_R$ is symmetric on the *weighted* space $L^2\big((0,\infty), \rho^{-1}\mathrm{d}\rho\big)$—the weight is the inverse lapse, inherited from the boost-energy norm, and tracking it is not pedantry: it is what decides the Weyl classification below. In the tortoise coordinate $x = \ln\rho \in \mathbb{R}$ the weight flattens, $\rho^{-1}\mathrm{d}\rho = \mathrm{d}x$, and

$$A_R \;\cong\; -\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \mu^2\mathrm{e}^{2x} \quad \text{on } L^2(\mathbb{R}, \mathrm{d}x). \tag{4}$$

The potential vanishes exponentially toward the horizon ($x \to -\infty$) and grows at $x \to +\infty$; both endpoints are at infinite distance with $V$ bounded below, hence both are limit point, and $A_R$ is essentially self-adjoint on $C_c^\infty(\mathbb{R})$ [9, Thm. X.8]. In the boost pair (boost clock, inverse-lapse weight) the horizon admits no boundary data: the Rindler instance of the tortoise folklore.

### 3.3 The adapted pair and the inverse-square foreshadowing

Transcribe the same fixed-frequency equation to the proper-distance variable $\rho$ with *flat* measure $\mathrm{d}\rho$—the pairing adapted to the infalling clock. Near the horizon (drop $\mu^2\rho^2$), $\psi = \rho^{\pm\mathrm{i}\omega}$, i.e. $\chi = \rho^{1/2}\psi \sim \rho^{1/2\pm\mathrm{i}\omega}$ solves

$$-\chi'' - \frac{\omega^2 + \frac{1}{4}}{\rho^2}\,\chi = 0, \tag{5}$$

as the indicial equation $s(s-1) = -(\omega^2 + \frac{1}{4})$, $s = \frac{1}{2} \pm \mathrm{i}\omega$, confirms. This is the inverse-square problem at super-critical (oscillatory) coupling for $\omega \neq 0$ ($\omega = 0$ is the marginal critical case, coupling $\frac{1}{4}$): a finite endpoint, limit-circle behavior, "fall to the center," and a $U(1)$ of boundary data—precisely the near-horizon conformal quantum mechanics of [19, 20, 21]. Same mode equation, different (weight, clock) pair, opposite Weyl verdict. Section 5.3 arbitrates.

### 3.4 The stochastic columns and the global dilation

Anticipating the Schwarzschild theorem, note what the wedge makes obvious. A diffusing particle in the wedge, watched in boost time, never reaches $\rho = 0$ (Theorem 5.3(a)): the horizon is a natural boundary, and the boost-time exterior dynamics is conservative on its own. Watched in inertial time, the same particle crosses in finite time; the exterior restriction loses mass through $\rho = 0$ (exit boundary in the generic regime $\kappa p D \leq v_0$, Theorem 5.3(b,c)). And here the conservative dilation of the lossy exterior semigroup is not hypothetical: it is literally the global Minkowski dynamics, with the "boundary data" being the state on the complementary wedge. The membrane-paradigm fork of Section 6—minimal lossy description versus completion by hidden degrees of freedom—is, in Rindler, the choice between the wedge-restricted process and the global one. What is a modeling fork at a black-hole horizon is, in flat space, resolved by the known global spacetime; the black-hole case differs in that no completion is canonically selected.

# 4 Schwarzschild: radial reduction, weights, and clocks

## 4.1 Reduction with the weight kept honest

On the exterior Schwarzschild geometry ($G = c = 1$)

$$\mathrm{d}s^2 = -f(r)\,\mathrm{d}t^2 + f(r)^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2, \qquad f(r) = 1 - \frac{2M}{r}, \tag{6}$$

with horizon $r_{\mathrm{h}} = 2M$ and surface gravity $\kappa = f'(r_{\mathrm{h}})/2 = 1/4M$, the Klein–Gordon equation $(\Box - m_\phi^2)\phi = 0$ separates as $\phi = \mathrm{e}^{-\mathrm{i}\omega t}\, Y_{\ell m}(\theta,\varphi)\, u(r)/r$ with

$$f\,\frac{\mathrm{d}}{\mathrm{d}r}\Big(f\,\frac{\mathrm{d}u}{\mathrm{d}r}\Big) + \big[\omega^2 - V_\ell(r)\big]u = 0, \qquad V_\ell(r) = f(r)\Big(\frac{\ell(\ell+1)}{r^2} + \frac{2M}{r^3} + m_\phi^2\Big), \tag{7}$$

the spin-zero Regge–Wheeler potential [7]. The mode dynamics in Killing time is $\partial_t^2 u = -A_\ell u$ with

$$A_\ell = -f\,\frac{\mathrm{d}}{\mathrm{d}r}\Big(f\,\frac{\mathrm{d}}{\mathrm{d}r}\Big) + V_\ell \qquad \text{on} \qquad \mathcal{H} = L^2\big((r_{\mathrm{h}},\infty),\, f(r)^{-1}\mathrm{d}r\big). \tag{8}$$

The weight $f^{-1}\mathrm{d}r$ is forced by symmetry of $A_\ell$ (equivalently by the conserved Killing energy): integrating by parts, $\langle u, A_\ell v\rangle_{\mathcal{H}} = \int f\,\bar{u}'v'\,\mathrm{d}r + \int V_\ell\,\bar{u}v\,f^{-1}\mathrm{d}r$ on compactly supported functions. We flag now, and use in Section 5.3, that this weight is exactly what moves the horizon endpoint between Weyl classes: it places the horizon at infinite $L^2$ distance, while the flat weight of the proper-distance variable places it at finite distance.

The tortoise coordinate $r_* = r + 2M\ln\big((r-2M)/2M\big)$, $\mathrm{d}r_* = \mathrm{d}r/f$, flattens the weight, $\mathcal{H} \cong L^2(\mathbb{R}, \mathrm{d}r_*)$, and

$$A_\ell \;\cong\; -\frac{\mathrm{d}^2}{\mathrm{d}r_*^2} + V_\ell(r(r_*)), \qquad V_\ell \sim C_\ell\,\mathrm{e}^{2\kappa r_*}\ (r_* \to -\infty), \qquad V_\ell \to m_\phi^2\ (r_* \to +\infty). \tag{9}$$

Near the horizon $f \simeq 2\kappa(r - r_{\mathrm{h}}) \simeq \text{const} \times \mathrm{e}^{2\kappa r_*}$: the lapse degenerates exponentially in the tortoise coordinate, at rate $2\kappa$.

## 4.2 Clocks and their time-of-flight integrals

An ingoing radial null ray has $\mathrm{d}r/\mathrm{d}t = -f$: the Killing-time flight from $r_0$ to the horizon is

$$T_{\mathrm{Killing}} = \int_{r_{\mathrm{h}}}^{r_0} \frac{\mathrm{d}r}{f(r)} \;\sim\; \frac{1}{2\kappa}\,\ln\frac{1}{r - r_{\mathrm{h}}} \;\longrightarrow\; \infty. \tag{10}$$

The Painlevé–Gullstrand (PG) time function $T = t + h(r)$, $h'(r) = \sqrt{2M/r}\,/f$, has unit lapse, regular horizon-crossing slices, and coincides with the proper time of observers falling from rest at infinity [8]; for such an observer $\mathrm{d}r/\mathrm{d}T = -\sqrt{2M/r}$ and

$$T_{\mathrm{PG}} = \int_{r_{\mathrm{h}}}^{r_0} \frac{\mathrm{d}r}{\sqrt{2M/r}} = \frac{2}{3\sqrt{2M}}\big(r_0^{3/2} - r_{\mathrm{h}}^{3/2}\big) < \infty. \tag{11}$$

Along the *ingoing null ray*, the PG clock rate relative to Killing time is

$$\frac{\mathrm{d}T}{\mathrm{d}t} = 1 + h'(r)\frac{\mathrm{d}r}{\mathrm{d}t} = 1 - \sqrt{\frac{2M}{r}} \;\simeq\; \frac{f}{2} \qquad (r \to r_{\mathrm{h}})\,: \tag{12}$$

a rate degenerating like the *first* power of the lapse function. The static observer's proper clock has rate $\sqrt{f}$ (half-power); hyperboloidal time functions [15, 17] are constructed precisely so that their rate along ingoing rays degenerates like $f$ and the flight is finite. Along the *outgoing* ray, by contrast, $\mathrm{d}T/\mathrm{d}t = 1 + \sqrt{2M/r} \to 2$: outgoing rays take infinite PG time to climb away from the horizon. The horizon splits the time-of-flight integral into an ingoing one (finite in adapted clocks) and an outgoing one (infinite in a genuine graph horizon-crossing clock, whose outgoing rate stays non-degenerate; the isotropic idealization of Definition 5.2 instead charges the outgoing leg at the same (degenerate) rate as the ingoing one, for a finite accumulated flight, which is exactly the source of its regular regime): this asymmetry is the trapping, and its noise-weighted counterpart is Feller's $\Sigma/N$ asymmetry in the exit regime—the signature of an exit boundary ($N$ itself becomes finite once the noise can climb the lapse, the regular case).

Interior continuation is not used in this paper except as the Rindler section's "global dilation" has an analogue: the PG slices continue smoothly to $r < r_{\mathrm{h}}$, and any dynamics continued across is one particular conservative completion of the exterior problem. The singularity then poses its own boundary problem [6, 30], which we do not touch.

# 5 The clock-dependence theorem

## 5.1 The stochastic model

We now prove the flagship in the column where it is theorem-grade. The model is the *radial transport diffusion*: the diffusion limit of a radially propagating exterior excitation subject to microscopic scattering.

**Definition 5.1** (exterior transport diffusion)**.** Fix $D > 0$ (noise strength) and $v_0 \geq 0$ (ingoing transport speed). In Killing time $t$ and the tortoise coordinate $x = r_* \in \mathbb{R}$, let

$$\mathrm{d}X_t = -v_0\,\mathrm{d}t + \sqrt{D}\,\mathrm{d}W_t. \tag{13}$$

The value $v_0 = 1$ is the transport value (radial characteristics move at unit tortoise speed, equation (10)); results are stated for general $v_0 > 0$, with the driftless case $v_0 = 0$ treated in Remark 5.6.

**Definition 5.2** (adapted clock)**.** An *adapted clock of lapse power* $p > 0$ is a time change $\tau(t) = \int_0^t \theta(X_s)\,\mathrm{d}s$ with rate $\theta : \mathbb{R} \to (0, \infty)$ continuous, bounded above, bounded away from zero on $[x_1, \infty)$ for some $x_1$, and

$$\theta(x) = c\,\mathrm{e}^{2\kappa p x}\big(1 + o(1)\big), \qquad x \to -\infty, \quad c > 0. \tag{14}$$

By (12) and the discussion following it: PG and hyperboloidal clocks have $p = 1$; the static proper clock has $p = \frac{1}{2}$ (a formal, interpolating time change—no static observer reaches the horizon, unlike the infalling PG family at $p = 1$); the Killing clock itself is $\theta \equiv 1$ ("$p = 0$"). This scalar time change is *isotropic*—$\theta$ depends on position alone, so ingoing and outgoing legs are charged at the same rate—and it reproduces the ingoing-null time-of-flight exactly; the genuine PG/hyperboloidal clock $T = t + h(r)$ is instead a graph functional whose rate differs on ingoing and outgoing legs (equation (12)), an anisotropy whose effect on the classification is taken up in Remark 5.7(iii). The time-changed process is $Y_\sigma = X_{\tau^{-1}(\sigma)}$, a regular diffusion with generator $\theta^{-1}\big(\frac{D}{2}\partial_x^2 - v_0\partial_x\big)$, unchanged scale function, and speed measure multiplied by $\theta$.

**Theorem 5.3** (clock dependence of the horizon boundary type)**.** *Let $X$ be as in Definition 5.1 with $v_0 > 0$, and let $Y$ be its time change under an adapted clock of lapse power $p > 0$. Then, for the horizon endpoint $x \to -\infty$:*

*(a)* (Killing clock) *The horizon is a* natural *boundary for $X$. In particular $X_t \to -\infty$ almost surely as $t \to \infty$ but $X$ never attains the horizon in finite time, the minimal semigroup is conservative, and no lateral (boundary) condition at the horizon is available: the Killing-time exterior dynamics is unique and conservative.*

*(b)* (adapted clock: attainability) *The total clock time $\tau_\infty = \int_0^\infty \theta(X_s)\,\mathrm{d}s$ is finite almost surely; $Y$ attains the horizon at the finite time $\tau_\infty$. Feller's attainability integral is* exactly *the ingoing time-of-flight of the clock: with $\Sigma$ as in (1),*

$$\Sigma(-\infty) = \frac{1}{v_0}\int_{-\infty}^{x_0} \theta(z)\,\mathrm{d}z = T_\tau^{\mathrm{in}}(x_0), \tag{15}$$

*the clock time accumulated along the ingoing characteristic $x(t) = x_0 - v_0 t$. For the Killing clock ($\theta \equiv 1$) this integral diverges, consistently with (a).*

*(c)* (adapted clock: type) *The horizon is an* exit *boundary for $Y$ if $\kappa p D \le v_0$ and a* regular *boundary if $\kappa p D > v_0$. In the exit case the minimal semigroup of $Y$ is strictly sub-stochastic, no lateral condition on the open exterior restores conservativity, and every conservative completion enlarges the state space by horizon degrees of freedom. In the regular case Feller's full menu of boundary conditions (Dirichlet/absorbing, Neumann/reflecting, elastic, sticky) opens at the horizon itself. The critical case $\kappa p D = v_0$ is exit by the classification above, and Proposition 6.1(i) (Dirichlet monotonicity) still applies there; but it sits outside the hypotheses of the universality and distinctness items, Proposition 6.1(ii)–(iii), which assume strict inequality on either side. What general wall data do* at *equality we leave open.*

*Proof.* The scale and speed of $X$ are $s'(x) = \mathrm{e}^{2v_0 x/D}$, so $s(x) = (D/2v_0)\,\mathrm{e}^{2v_0 x/D}$ with $s(-\infty) = 0$, and $m'(x) = (2/D)\,\mathrm{e}^{-2v_0 x/D}$. Time change leaves $s$ fixed and multiplies the speed density by $\theta$: $m'_Y(x) = (2/D)\,\theta(x)\,\mathrm{e}^{-2v_0 x/D}$ [11, 12].

(a) Take $\theta \equiv 1$. In (1), $\mathcal{M}(x_0) - \mathcal{M}(\xi) = \int_\xi^{x_0} (2/D)\mathrm{e}^{-2v_0 z/D}\mathrm{d}z \sim (1/v_0)\mathrm{e}^{-2v_0\xi/D}$ as $\xi \to -\infty$, while $\mathrm{d}s(\xi) = \mathrm{e}^{2v_0\xi/D}\mathrm{d}\xi$; the integrand of $\Sigma$ tends to the constant $1/v_0$ and $\Sigma = \infty$. Likewise $s(x_0) - s(\eta) \to s(x_0) > 0$ while $\mathrm{d}\mathcal{M}(\eta) = (2/D)\mathrm{e}^{-2v_0\eta/D}\mathrm{d}\eta$ diverges, so $N = \infty$. Both integrals infinite: natural. That $X_t \to -\infty$ a.s. is the strong law for (13); unattainability in finite time and conservativity of the minimal semigroup are the standard consequences of the natural classification, and Feller's lateral-condition theory attaches conditions only at accessible boundaries [10, 11].

(b) By the strong law there is an a.s.-finite $T(\omega)$ with $X_s \le -v_0 s/2$ for $s \ge T$; hence $\theta(X_s) \le C\mathrm{e}^{-\kappa p v_0 s}$ eventually and $\tau_\infty < \infty$ a.s. Since $X_t \to -\infty$ while $\tau(t) \uparrow \tau_\infty$, the time-changed path $Y_\sigma$ converges to the horizon as $\sigma \uparrow \tau_\infty$: the boundary is attained at the finite time $\tau_\infty$, almost surely. The identity (15) is Tonelli (nonnegative integrand):

$$\Sigma(-\infty) = \int_{-\infty}^{x_0} \mathrm{d}\mathcal{M}_Y(z) \int_{-\infty}^{z} \mathrm{d}s(\xi) = \int_{-\infty}^{x_0} \frac{2\theta(z)}{D}\,\mathrm{e}^{-2v_0 z/D} \cdot \frac{D}{2v_0}\,\mathrm{e}^{2v_0 z/D}\,\mathrm{d}z = \frac{1}{v_0}\int_{-\infty}^{x_0} \theta(z)\,\mathrm{d}z,$$

finite for every adapted clock (exponentially decaying $\theta$), and equal to the clock time along the ingoing characteristic, for which $\mathrm{d}\tau = \theta\,\mathrm{d}t = \theta\,|\mathrm{d}x|/v_0$.

(c) The entrance integral is

$$N(-\infty) = \frac{1}{v_0} \int_{-\infty}^{x_0} \theta(\eta) \left( \mathrm{e}^{2v_0(x_0-\eta)/D} - 1 \right) \mathrm{d}\eta, \tag{16}$$

whose integrand behaves as const $\times\, \mathrm{e}^{2(\kappa p - v_0/D)\eta}$ for $\eta \to -\infty$: divergent iff $\kappa p \le v_0/D$. With $\Sigma < \infty$ always, the classification (1) gives exit iff $\kappa p D \le v_0$ and regular iff $\kappa p D > v_0$. In the exit case, strict sub-stochasticity is (b): $P_\sigma^{\min}\mathbf{1}(x) = \mathbb{P}_x[\tau_\infty > \sigma] < 1$ for $\sigma > 0$; the statements on lateral conditions and on conservative completions (absorption at an adjoined cemetery state, or re-injection mechanisms, all involving the boundary point as a new state) are Feller's exit-boundary theory [10, 11]. □

**Corollary 5.4** (mass–flux identity: the membrane's absorption law)**.** *In the exit regime, the minimal semigroup's deficit is the horizon flux:*

$$1 - P_\sigma^{\min}\mathbf{1}(x) = \mathbb{P}_x\big[\tau_\infty \le \sigma\big], \qquad -\frac{\mathrm{d}}{\mathrm{d}\sigma} P_\sigma^{\min}\mathbf{1}(x) = \Phi_{\mathrm{h}}(\sigma; x), \tag{17}$$

*where $\Phi_{\mathrm{h}}(\sigma; x)$ is the probability flux through the horizon at clock time $\sigma$. Nothing is "lost": the deficit is accounted, to the exterior observer, as absorption by the boundary—the defining bookkeeping of the membrane paradigm (Section 6).*

*Remark* 5.5 (the two regimes and the stretched-horizon skin depth). The exit condition $\kappa p D \le v_0$ says: the noise's upstream penetration depth $D/v_0$ (the distance over which diffusion can fight the ingoing transport) does not exceed the lapse e-folding length $1/\kappa p$ of the clock rate.[1] For a macroscopic horizon, $1/\kappa = 4M$ is macroscopic while $D/v_0$ is set by microphysics: deep exit regime; the horizon is a one-way membrane and no boundary data exist even in the adapted clock. The opposite regime $\kappa p D > v_0$—noise that can climb an e-folding of the lapse—renders the horizon *regular*: attainable *and* enterable, with a full family of boundary conditions (this regular regime is a feature of the isotropic clock idealization and does not survive a genuine graph clock; see Remark 5.7(iii)). This is the stochastic twin of the near-horizon limit-circle regime of conformal quantum mechanics [19, 20, 18] (cf. Sections 3.3 and 5.3), and it is exactly the regime in which brick-wall data survive wall removal (Proposition 6.1). The classification is decided by *two* time-of-flight integrals—ingoing ($\Sigma$, equation (15)) and outgoing ($N$, equation (16), the clock time of upstream excursions weighted *inversely* by their gambler's-ruin rarity $\mathrm{e}^{-2v_0\Delta x/D}$, i.e. by $\mathrm{e}^{2v_0\Delta x/D}$)—which coincide in the symmetric half-line problems of [1] but split at a horizon. The split *is* the trapping.

*Remark* 5.6 (driftless clock change: a white-hole boundary). For $v_0 = 0$ (pure noise, no transport) the same computation gives $\Sigma = \infty$, $N < \infty$ under any adapted clock: an *entrance* boundary—unattainable from outside but a legitimate starting point. This driftless case is marginal; a genuine white-hole horizon carries *outward* drift ($v_0 < 0$, the mirrored model: Definition 5.1 fixes $v_0 \ge 0$), which likewise makes the boundary entrance. Exit and entrance are exchanged by time reversal, as they should be: the classification tracks the transport structure, not the geometry alone.

[1] Both lengths are quoted without the factor of 2 carried by the exponents of the scale density and clock rate ($s' \propto \mathrm{e}^{2v_0 x/D}$, $\theta \propto \mathrm{e}^{2\kappa p x}$); the natural $e$-folding lengths are $D/2v_0$ and $1/2\kappa p$. The factor cancels in the ratio, so the exit inequality $\kappa p D \le v_0$ is unaffected.

*Remark* 5.7 (robustness and the shift caveat). Two honesty notes. (i) The theorem is stated for the model of Definition 5.1, but the classification depends only on the near-horizon asymptotics of scale and speed: any diffusion whose Killing-time coefficients in the tortoise coordinate are bounded, uniformly elliptic, and asymptotically constant near $r_* = -\infty$ yields the same conclusions with $v_0, D$ read off from the limits. (ii) A scalar time change is the exact transported dynamics only for clock functions constant on the transported slices; the PG slicing in addition tilts the slices (the same Killing field generates the evolution; the *slicing* changes—see Section 5.4), which in the reduced dynamics contributes an additional bounded *inward* drift near the horizon. Inspection of (15) and (16) shows both integrals are monotone in the ingoing drift in the direction of attainability and exit: the neglected term can only reinforce the exit verdict. So the exit half is robust—macroscopic horizons sit deep in $\kappa pD \ll v_0$—whereas the *regular* regime near the knife-edge $\kappa pD \approx v_0$ is the fragile side: the same neglected inward drift can push it across into exit, so the boundary-data-survive conclusion is the delicate one and is offered as such. We flag rather than formalize this, and correspondingly attribute to the theorem only the time-change statement. (iii) The time change of Definition 5.2 is isotropic, whereas the genuine graph clock $T = t + h(r)$ is not a scalar time change at all: along the diffusion $\mathrm{d}T = \mathrm{d}t + h'(r)\,\mathrm{d}X$ mixes drift and noise, and on *outgoing* (upstream) legs its rate does not degenerate (outgoing rays take infinite adapted time to climb away from the horizon; see the discussion at (12)). The one robust consequence is that a genuine clock charges each closed upstream excursion its full, non-degenerate duration—the $h$-terms cancel over a round trip—so the analogue of the entrance integral (16) diverges and no boundary data open at the horizon: the regular regime $\kappa pD > v_0$ is thus specific to the isotropic idealization and does not survive an honest graph clock, consistent with the paper's own observation that a genuine graph clock's outgoing time-of-flight does not degenerate. We do *not* read a sharp attainability verdict off the honest functional—at the transport value $v_0 = 1$ it is not even monotone in $t$, and a faithful treatment is the slicing-level SDE with its tilt drift (ii), or the relativistic diffusion of [22], neither carried out here. What we retain is the negative statement: boundary-data survival (the regular regime) is a model artifact, consistent with its fragile status in Proposition 6.1(iii).

*Remark* 5.8 (relation to relativistic diffusions). The genuinely relativistic diffusion on Schwarzschild (white-noise-in-proper-time acceleration) was constructed by Franchi and Le Jan [22], who prove capture by the black hole in finite proper time—the attainability half of our exit verdict, for a process with the causal structure built in. A Feller boundary classification of the horizon for their process, and the natural/exit flip as a parameterization statement, do not appear there or, to our knowledge, anywhere; Theorem 5.3 is the transport-diffusion version, and extending it to the Franchi–Le Jan process is listed in Section 8.

### 5.2 The quantum column: proved and imported

**Proposition 5.9** (Killing pair: essential self-adjointness)**.** *For every $\ell$ and $m_\phi \geq 0$, the operator $A_\ell$ of (8) is essentially self-adjoint on $C_c^\infty(r_\mathrm{h}, \infty)$; equivalently (9) is essentially self-adjoint on $C_c^\infty(\mathbb{R})$. The Killing-time mode dynamics is the unique conservative dynamics; the Wald–Ishibashi–Wald prescription [3, 4] is available and trivial (the positive self-adjoint extension is unique); no horizon boundary data exist. The same holds for the Rindler operator (4).*

*Proof.* The unitary $\mathcal{H} \to L^2(\mathbb{R}, \mathrm{d}r_*)$ is the identification along $\mathrm{d}r_* = \mathrm{d}r/f$. In (9) the potential is continuous and bounded below (by 0) with both endpoints at infinite coordinate distance; the classical Weyl/Sears limit-point criteria apply at both ends [9, Thm. X.8 and Appendix to

X.1]. □

This is folklore—stated in passing throughout the quantum-singularity and quasinormal-mode literatures (e.g. [6])—and is included only because the proof is shorter than a citation trail. The point of dwelling on it is the pairing: limit point holds *in the Killing pair* (Killing clock, inverse-lapse weight $f^{-1}\mathrm{d}r$), where the horizon sits at infinite $L^2$ distance.

**Proposition 5.10** (adapted pair: imported)**.** *On horizon-crossing (hyperboloidal) slicings, the generator of the exterior mode evolution is non-self-adjoint on the natural energy Hilbert space; it generates a strongly continuous semigroup whose spectrum contains the quasinormal frequencies as genuine eigenvalues, and its non-normality is associated with energy loss through the horizon (and through null infinity). This is Warnick's theorem [15] (established for asymptotically anti-de Sitter black holes; the asymptotically-flat generator-eigenvalue statement, with loss through null infinity, is Gajic–Warnick [16], in a model problem) and the pseudospectrum analysis of [17]; we import it as the quantum statement of the adapted-clock column and re-derive none of it.*

The division of labor between Propositions 5.9 and 5.10 is deliberate and should be read as a caveat: for the *diffusion*, the passage between clocks is an honest time change and Theorem 5.3 is a theorem; for the *field*, the passage between the static and hyperboloidal descriptions involves the energy norms of the hyperboloidal program, and the clock-dependence statement at QFT strength is exactly as strong as the cited results—no stronger. The dictionary claim of this paper is that the two columns exhibit the same structure, with the stochastic column proved and the quantum column cited.

### 5.3 Arbitrating limit point versus limit circle at the horizon

The tension advertised in the introduction can now be discharged. The tortoise-coordinate folklore (limit point, no data) and the near-horizon conformal-QM literature (limit circle, $U(1)$ of data) are talking about *different pairs*:

| | Killing pair | adapted pair |
|---|---|---|
| weight | $f^{-1}\mathrm{d}r$ (inverse lapse) | flat ($\mathrm{d}\rho$, proper-distance-like) |
| horizon $L^2$ distance | infinite | finite |
| fixed-$\omega$ operator | $-\partial_{r_*}^2 + V$, $V \to 0$ | inverse-square, coupling $> \frac{1}{4}$ ($\omega \neq 0$) |
| Weyl verdict | limit point | limit circle (oscillatory; flat-weight, see text) |
| time-of-flight | infinite (10) | finite (11) |
| Feller verdict (Thm. 5.3) | natural | exit (regular if $\kappa pD > v_0$) |
| boundary data | none possible | $U(1)$ (Weyl, LC); none/menu (Feller: exit/regular) |

One asymmetry between the columns should be read off the table explicitly: the exit/regular refinement is a *Feller-column* phenomenon, keyed to $\kappa pD$ versus $v_0$; the Weyl column has no such split—the adapted-pair operator is limit circle (oscillatory) for every $\omega \neq 0$ irrespective of that ratio—so the "exit"/"regular" annotations in the boundary-data row are the stochastic reading, not a quantum dichotomy—and the "regular" entry is itself a feature of the isotropic clock idealization that a genuine graph clock removes (Remark 5.7(iii)). The two columns share the natural/Killing-pair verdict and the finite-versus-infinite time-of-flight that drives everything; they need not agree on the finer lateral structure, and the paper's "same structure" claim is at the level of that shared decider, not of the exit/regular subdivision. Each column is internally consistent, and the dictionary's decider predicts each verdict from that column's time-of-flight

integral. The near-horizon truncation of [19, 20] works in the adapted pair, where the horizon is at finite distance and data *can* live—in the stochastic column, precisely in the regular regime of Remark 5.5. One detail of [18] deserves emphasis because it anticipates Section 6: their deficiency indices $(1, 1)$ and $U(1)$ family arise on a truncated near-horizon region $(-\infty, x_{\max}]$, with the extension living at the *outer edge* $x_{\max}$—a stretched-horizon wall—while the horizon end itself remains limit point. The boundary data of the near-horizon literature are wall data, not horizon data; the dictionary will say the same thing probabilistically.

### 5.4 States versus domains; which "clock"

Two further honesty caveats, kept separate from the theorems.

*States versus domains.* The Boulware/Unruh/Hartle–Hawking choices, and the Kay–Wald uniqueness theorem for the Hartle–Hawking state on the bifurcate horizon [31], concern boundary data at the level of *states* (two-point functions, positive frequency splits). Everything in this paper lives at the level of *generators and domains*—which dynamics, not which state. The two axes are independent: the Killing-pair dynamics is unique (Proposition 5.9) while the state choice on it remains free; conversely the adapted-clock completions of Section 6 are dynamics choices made before any state is selected. Conflating the axes is a standard referee objection to horizon-boundary claims, and nothing below trades on it.

*Which clock.* PG time is generated by the *same* Killing field as Schwarzschild time: the exterior is stationary either way, and "changing the clock" is not a change of coordinates on a fixed problem but a change of the pair (space of data on slices, parameterization of evolution)—in the stochastic column, of the parameterization of the transported process, which is where Theorem 5.3 lives; in the quantum column, of the (Hilbert space, slicing) pair, which is where Proposition 5.10 lives. The invariant content of S1 is: the Feller/Weyl classification is a property of the pair, not of the geometry; the geometry supplies only the two time-of-flight integrals that decide each pair's verdict.

## 6 The membrane paradigm as the minimal process

### 6.1 The membrane paradigm, restated

The membrane paradigm [23, 24, 25] replaces the black-hole interior, for exterior purposes, by a fictitious dissipative fluid on a stretched horizon: a timelike surface just outside $r_{\mathrm{h}}$ endowed with Ohmic resistivity, shear and (negative) bulk viscosity, and absorptivity, tuned so that exterior physics is reproduced. Parikh and Wilczek derived it from a variational principle: discarding the interior ruins the action's stationarity, and a surface term on the stretched horizon must be added to restore it—dissipation as the price of truncation. Susskind, Thorlacius, and Uglum [26] promoted the stretched horizon to the carrier of the complementarity degrees of freedom.

In the dictionary, "truncate the interior and pay a dissipative surface term" has an exact translation: pass to the exterior restriction of the dynamics, i.e. to the *minimal semigroup* of the exterior problem in a horizon-adapted clock. By Theorem 5.3(b,c) this object is sub-stochastic, contractive, and its deficit is precisely the horizon flux (Corollary 5.4): the membrane's absorption law. The identification to be defended in this section is:

> *membrane paradigm = minimal sub-stochastic semigroup of the exterior problem in horizon-adapted time; unitary completions = conservative dilations, parameterized*

*by data at a stretched-horizon wall; the minimal object, and only it, is regulator-independent.*

This identification is theorem-grade for the radial diffusion $Y$—Proposition 6.1 and Corollary 5.4 establish it there—while for the field it is a dictionary reading carried by the same operator data, not a separate theorem.

### 6.2 Walls, completions, and the wall-removal limit

A stretched horizon at $r = r_{\mathrm{h}} + \epsilon$ is, in the tortoise coordinate, a wall at $x = -R$ with $R \simeq (2\kappa)^{-1} \ln(1/\epsilon) \to \infty$: an ordinary interior point promoted to a boundary. For the truncated problem on $[-R, \infty)$ the wall is a *regular* boundary, so Feller's full menu of lateral conditions is available there—absorbing (stochastic Dirichlet), reflecting (Neumann), elastic, sticky. A word on "Dirichlet," which flips sense between the columns: 't Hooft's brick wall [27] imposes $\phi = 0$ on the *field*, a flux-free (hence conservative, unitarity-preserving) reflecting box, whose stochastic counterpart is the *reflecting* (Neumann) condition; the stochastic *Dirichlet* (absorbing) condition, by contrast, kills the process and builds the minimal semigroup (Proposition 6.1(i)), and is not a conservative completion at all. In the quantum column, the $U(1)$ extension family at the wall of [18], and, in multichannel form, S-matrix identifications of which 't Hooft's antipodal map [28] is the structural example (a channel-scrambling unitary at the boundary, in the completion menu of [1]). Each conservative wall condition defines a conservative dynamics: a dilation of the exterior evolution carrying boundary data. The Rindler wedge (Section 3) shows the construction is not exotic: there, one particular dilation is the global Minkowski dynamics, with the second wedge as the "wall" degrees of freedom; on Schwarzschild, continuation through the horizon on PG slices plays the same role. Abstract dilation theory [14, 13] guarantees a unitary dilation *exists* (the minimal one is essentially unique); the *multiplicity* of conservative completions—which boundary data—is not a dilation-theoretic freedom but the family of lateral (self-adjoint-extension) conditions at the regular wall. The physical question is which member the horizon selects.

**Proposition 6.1** (the minimal object and wall removal)**.** *Consider the adapted-clock process $Y$ of Theorem 5.3 with walls at $-R \to -\infty$; the boundary regime is stated per item (exit $\kappa p D \le v_0$ for (i)–(ii), strict where indicated; regular $\kappa p D > v_0$ for (iii)).*

*(i)* (Dirichlet monotonicity) *The Dirichlet-wall semigroups increase monotonically to the minimal semigroup as $R \to \infty$, independently of the regularization sequence: $P_\sigma^{(R),\mathrm{Dir}} \uparrow P_\sigma^{\min}$.*

*(ii)* (wall-removal universality for local conditions, exit regime, $\kappa p D < v_0$) *Every* local *lateral condition converges to the same limit. Quantitatively, for the reflecting wall: the expected clock time for $Y$ started at the wall to return to a fixed exterior level $x_0$ diverges as*

$$\mathbb{E}_{-R}\big[T_{x_0}\big] = \int_{-R}^{x_0} \big[\mathcal{M}_Y(\xi) - \mathcal{M}_Y(-R)\big]\,\mathrm{d}s(\xi) \;\asymp\; \mathrm{e}^{\,2(v_0-\kappa p D)R/D} \;\longrightarrow\; \infty, \tag{18}$$

*and for every fixed clock horizon $\sigma$ and compact exterior set, the probability that a wall-touching path re-enters by time $\sigma$ is $O\big(\sigma\,\mathrm{e}^{-2(v_0-\kappa p D)R/D}\big)$; since every* local *lateral condition is dominated between the Dirichlet (absorbing) and reflecting (Neumann) extremes, all such wall dynamics converge to $P^{\min}$ on exterior observables as $R \to \infty$. The domination sandwich is what fails for non-local (Feller–Wentzell jump/re-injection) conditions: a wall*

*that re-injects the lost mass at a fixed exterior point is not dominated by the reflecting extreme and keeps the dynamics conservative, so the universality is asserted for the local menu only. This is the probabilistic counterpart of the abstract's "rigorously for the $U(1)$ family, conjecturally in the multichannel (antipodal) case" hedge: the channel-scrambling (antipodal) boundary conditions are the non-local ones the sandwich does not reach.*

*(iii)* (regular regime) *If $\kappa p D > v_0$, return times from the wall remain bounded as $R \to \infty$, distinct wall conditions retain distinct limits, and boundary data survive wall removal: the brick-wall family becomes a genuine horizon family. (This regular regime is the fragile side of the classification; see the tilt-drift caveat and Remark 5.7(iii), where a genuine (anisotropic) graph clock removes this regime altogether. Its distinctness claim is at the same sketch strength as (ii).)*

*Proof (with two steps at sketch strength).* (i) is the standard construction of the minimal semigroup as the monotone limit of killed semigroups on an exhausting sequence of subdomains [11, 13]. (ii): the displayed formula is the classical expected-hitting-time formula for a reflected diffusion [12, Ch. 15]; with $m'_Y(\xi) \asymp \mathrm{e}^{2(\kappa p - v_0/D)\xi}$ and $\mathrm{d}s(\xi) = \mathrm{e}^{2v_0\xi/D}\mathrm{d}\xi$ the stated growth follows. For the re-entry probability, decompose the reflected path at the wall into excursions crossing $[-R, -R+1]$: each excursion reaches $x_0$ before returning to the wall with probability $p_R = [s(-R+1) - s(-R)]/[s(x_0) - s(-R)] \asymp \mathrm{e}^{-2v_0 R/D}$ (gambler's ruin in scale), while the mean excursion duration is $\mu_R \asymp \mathrm{e}^{-2\kappa p R}$ (speed density times scale increment near the wall), so the expected number of excursions by clock time $\sigma$ is $O(\sigma/\mu_R)$ and $\mathbb{P}[\text{re-entry by } \sigma] = O(\sigma p_R/\mu_R) = O\big(\sigma \mathrm{e}^{-2(v_0 - \kappa p D)R/D}\big)$; the constants, and the domination of intermediate lateral conditions between the absorbing and reflecting extremes, are standard but not spelled out here, and we grade this step a sketch. (iii): the same formulas with $\kappa p D > v_0$ give bounded return times and non-vanishing re-entry probabilities; distinctness of the limits follows from distinctness at any fixed wall together with the uniform non-degeneracy of the re-entry channel—graded a sketch at the same strength as (ii). □

### 6.3 The dictionary of completions

| membrane-paradigm object | dictionary object |
|---|---|
| stretched horizon at $r_\mathrm{h} + \epsilon$ | regular wall at tortoise depth $R \sim \frac{1}{2\kappa} \ln \frac{1}{\epsilon}$ |
| membrane dissipation (Ohmic absorption) | sub-stochastic deficit = horizon flux (Cor. 5.4) |
| Parikh–Wilczek surface term | generator's boundary bookkeeping after truncation |
| brick wall (field Dirichlet = reflecting box) [27] | reflecting (Neumann) condition: a conservative dilation |
| stochastic Dirichlet (absorbing) wall | $P^{\min}$-approximant (Prop. 6.1(i)), not a completion |
| near-horizon $U(1)$ family [18, 19] | elastic/mixed conditions (wall extension family) |
| antipodal S-matrix [28] | channel-scrambling unitary boundary condition |
| complementarity DOF [26] | dilation (boundary) degrees of freedom |
| interior continuation (PG/Kruskal) | one particular conservative dilation |
| membrane paradigm itself | the minimal sub-stochastic semigroup |
| wall-position independence | exit property + Prop. 6.1(ii) |

### 6.4 Scattering equivalence, entropy inequivalence

Balachandran, de Queiroz, and Saa proved that their $U(1)$ of wall extensions are mutually equivalent for exterior scattering and for thermal response [18]: exterior S-matrix observables do

not see the extension parameter. Proposition 6.1(ii) is the diffusion analogue and a structural extension of that lemma: in the exit regime, *all local* wall completions—not only the $U(1)$ family—collapse onto the minimal semigroup for exterior observables as the wall is removed (non-local re-injection completions excepted, as in the proposition). Exterior scattering and thermality are properties of the minimal object; the completions decorate it invisibly.

Entropy is different, and here we claim no theorem—only a reading of cited results. The brick-wall entropy diverges as the wall approaches the horizon and depends on the wall position [27]; the near-horizon extension parameter feeds the entropy in [19, 20, 21]; and [18] frames the family as a brick-wall generalization. In the dictionary this pattern is forced: entropy counts *boundary* (dilation) states, which are exactly the states that do not survive the wall-removal limit (Proposition 6.1(ii): the re-entry channel closes; the boundary states decouple from exterior dynamics but are still counted). Entropy is a property of the chosen dilation, not of the minimal object; extension dependence of horizon entropy is then the expected behavior, and the brick wall's notorious cutoff dependence is extension dependence in the Wald–Ishibashi–Wald sense [3, 4]. The divergence as $\epsilon \to 0$ is, in this reading, the price of insisting on a conservative exterior: the dilation must supply ever more boundary states as its wall chases a boundary that—in the exit regime—can carry none.

## 7 The information paradox as a dictionary entry

This section is a discussion, fenced as follows: fixed Schwarzschild background; no evaporation, no backreaction, no holography; all statements concern the exterior algebra of a test field and its diffusion analogue; "information" means conservativity (stochasticity/unitarity) of the exterior dynamics, not fine-grained entropy of radiation. Within that fence, the dictionary contributes one structural sentence to the paradox: *"does the horizon need extra data?" is a clock-indexed question, and its two answers are not in tension because the boundary classification they refer to is not clock-invariant* (Theorem 5.3).

In Killing time the horizon is natural: no data can be imposed, the exterior dynamics is unique and conservative (Proposition 5.9), and the distant observer's unitarity involves no choice whatsoever. This is the mathematically precise core of the "no drama at the level of exterior dynamics" intuition, and of complementarity's outside story [26]: exterior unitarity is not an assumption to be defended but the unique option in that pair.

In infalling time the horizon is generically exit: the choice-free description is the minimal, sub-stochastic one—the membrane (Section 6)—and it is *lossy for the exterior algebra* by theorem, not by approximation. An exterior observer who insists on conservativity in this clock must complete the dynamics, and every completion installs boundary data (dilation degrees of freedom) at a stretched horizon: brick wall, $U(1)$ extension, antipodal S-matrix, or interior continuation. The data are optional (refusing them is the membrane), underdetermined (inequivalent choices exist), and exterior-scattering-invisible (Section 6.4)—which is why the debate about them is conducted in entropy accounting rather than in S-matrix elements.

The firewall argument [29] sharpens a contradiction among assumptions that include exterior unitarity and interior regularity for the infalling observer. The dictionary does not adjudicate the quantum-gravitational content of that argument, and nothing here claims to. What it does say is narrower: *insofar as* the argument presupposes a single, slicing-invariant verdict on what the horizon carries, that presupposition is unavailable already at the level of a test field's boundary classification—the Killing-clock answer ("nothing, necessarily") and the infalling-clock answer

("either loss or data, your choice") are both correct and refer to different (clock, Hilbert-space) pairs. To the extent a tension rests on that presupposition, it is relieved once the pair is specified and is not a property of the horizon itself. Whether the firewall argument in fact rests on it is a quantum-gravity question this dictionary does not settle.

Two boundary-data proposals sit outside this discussion and are flagged to prevent conflation: the Horowitz–Maldacena final-state condition [30] installs data at the *singularity*, the interior's own boundary problem [6], which the exterior dictionary does not reach; and vacuum choice (Boulware/Unruh/Hartle–Hawking, [31]) is state selection, orthogonal to every domain statement made here (Section 5.4).

# 8 Outlook

Four continuations are concrete. *(1) Analogue horizons.* The acoustic horizon of a transonic flow [32, 33] is a laboratory half-line problem in which the two regimes of Remark 5.5 are physical dials: subluminal dispersion regularizes the horizon into an effectively enterable (regular-like) boundary—mode conversion replacing boundary data [34]—while dissipative near-horizon physics selects the minimal semigroup. A Feller classification of acoustic horizons, with the dispersive scale playing $D/v_0$ against the analogue surface gravity, is directly testable in principle in BEC platforms. *(2) The QFT upgrade of S1.* Promoting the adapted-clock column from cited (Proposition 5.10) to proved requires the hyperboloidal energy norms of [15, 17]; the target is a theorem equating the Feller time-change statement with a similarity/domain statement for the field generator pair. *(3) Relativistic diffusions.* Extending Theorem 5.3 to the Franchi–Le Jan process [22], whose causal structure enforces the exit regime without a model parameter, would remove Definition 5.1's modeling freedom. *(4) Rotation.* Kerr adds superradiance—an energy flux with a sign structure the sub-stochastic column does not yet have—and the near-horizon geometry of extremality, where $\kappa \to 0$ degenerates the e-folding length and the two-regime boundary $\kappa pD = v_0$ collapses. Evaporation and backreaction remain outside the program's scope until the fixed-background dictionary is exhausted.

The summary sentence of the paper: a horizon is not a boundary *simpliciter*; it is a boundary relative to a clock, and one time-of-flight integral per clock—two once trapping splits them—prices everything: the loss of self-adjointness, the availability of boundary data, the sub-stochastic deficit of the membrane, and the entropy bill of every attempt to refuse it.


# Acknowledgments

The framework applied here is that of the companion papers [1, 2]; the debts to Wald and Ishibashi–Wald, to the hyperboloidal program, and to the near-horizon extension literature are recorded in Section 1.3.


# References


[1] J. Torkelson, *Indivisible stochastic processes for non-Hermitian and higher-derivative quantum systems: detailed balance, broken $\mathcal{PT}$ symmetry, and the explosion of ghosts*, companion paper (2026), https://github.com/embeddetech/p01-indivisible-ghosts.

[2] J. Torkelson, *Ghost vacuum decay as a stochastic process: instantaneous states, the boost-orbit measure, and the foliation that ghosts make visible*, companion paper (2026), https://github.com/embeddetech/p03-instantaneous-ghosts.

[3] R. M. Wald, *Dynamics in nonglobally hyperbolic, static space-times*, J. Math. Phys. **21**, 2802 (1980).

[4] A. Ishibashi and R. M. Wald, *Dynamics in non-globally-hyperbolic static space-times II: general analysis of prescriptions for dynamics*, Class. Quantum Grav. **20**, 3815 (2003), arXiv:gr-qc/0305012.

[5] A. Ishibashi and R. M. Wald, *Dynamics in non-globally-hyperbolic static space-times III: anti-de Sitter space-time*, Class. Quantum Grav. **21**, 2981 (2004), arXiv:hep-th/0402184.

[6] G. T. Horowitz and D. Marolf, *Quantum probes of spacetime singularities*, Phys. Rev. D **52**, 5670 (1995), arXiv:gr-qc/9504028.

[7] T. Regge and J. A. Wheeler, *Stability of a Schwarzschild singularity*, Phys. Rev. **108**, 1063 (1957).

[8] K. Martel and E. Poisson, *Regular coordinate systems for Schwarzschild and other spherical spacetimes*, Am. J. Phys. **69**, 476 (2001), arXiv:gr-qc/0001069.

[9] M. Reed and B. Simon, *Methods of Modern Mathematical Physics II: Fourier Analysis, Self-Adjointness*, Academic Press, New York (1975).

[10] W. Feller, *The parabolic differential equations and the associated semi-groups of transformations*, Ann. of Math. **55**, 468 (1952).

[11] K. Itô and H. P. McKean, Jr., *Diffusion Processes and their Sample Paths*, Springer, Berlin (1965).

[12] S. Karlin and H. M. Taylor, *A Second Course in Stochastic Processes*, Academic Press, New York (1981).

[13] E. B. Davies, *One-Parameter Semigroups*, Academic Press, London (1980).

[14] B. Sz.-Nagy and C. Foiaş, *Harmonic Analysis of Operators on Hilbert Space*, North-Holland, Amsterdam (1970).

[15] C. M. Warnick, *On quasinormal modes of asymptotically anti-de Sitter black holes*, Commun. Math. Phys. **333**, 959 (2015), arXiv:1306.5760.

[16] D. Gajic and C. M. Warnick, *A model problem for quasinormal ringdown of asymptotically flat or extremal black holes*, J. Math. Phys. **61**, 102501 (2020), arXiv:1910.08481.

[17] J. L. Jaramillo, R. Panosso Macedo, and L. Al Sheikh, *Pseudospectrum and black hole quasinormal mode (in)stability*, Phys. Rev. X **11**, 031003 (2021), arXiv:2004.06434.

[18] A. P. Balachandran, A. R. de Queiroz, and A. Saa, *Near-horizon modes and self-adjoint extensions of the Schrödinger operator*, in *Classical and Quantum Physics*, Springer Proc. Phys. **229** (2019), arXiv:1810.00020.

[19] K. S. Gupta and S. Sen, *Further evidence for the conformal structure of a Schwarzschild black hole in an algebraic approach*, Phys. Lett. B **526**, 121 (2002), arXiv:hep-th/0112041.

[20] H. E. Camblong and C. R. Ordóñez, *Black hole thermodynamics from near-horizon conformal quantum mechanics*, Phys. Rev. D **71**, 104029 (2005), arXiv:hep-th/0411008.

[21] H. E. Camblong and C. R. Ordóñez, *Conformal enhancement of holographic scaling in black hole thermodynamics: a near-horizon heat-kernel framework*, JHEP **0712**, 099 (2007), arXiv:0709.2942.

[22] J. Franchi and Y. Le Jan, *Relativistic diffusions and Schwarzschild geometry*, Comm. Pure Appl. Math. **60**, 187 (2007), arXiv:math/0410485.

[23] T. Damour, *Black-hole eddy currents*, Phys. Rev. D **18**, 3598 (1978).

[24] K. S. Thorne, R. H. Price, and D. A. Macdonald (eds.), *Black Holes: The Membrane Paradigm*, Yale University Press, New Haven (1986).

[25] M. Parikh and F. Wilczek, *An action for black hole membranes*, Phys. Rev. D **58**, 064011 (1998), arXiv:gr-qc/9712077.

[26] L. Susskind, L. Thorlacius, and J. Uglum, *The stretched horizon and black hole complementarity*, Phys. Rev. D **48**, 3743 (1993), arXiv:hep-th/9306069.

[27] G. 't Hooft, *On the quantum structure of a black hole*, Nucl. Phys. B **256**, 727 (1985).

[28] G. 't Hooft, *Black hole unitarity and antipodal entanglement*, Found. Phys. **46**, 1185 (2016), arXiv:1601.03447.

[29] A. Almheiri, D. Marolf, J. Polchinski, and J. Sully, *Black holes: complementarity or firewalls?*, JHEP **02** (2013) 062, arXiv:1207.3123.

[30] G. T. Horowitz and J. Maldacena, *The black hole final state*, JHEP **02** (2004) 008, arXiv:hep-th/0310281.

[31] B. S. Kay and R. M. Wald, *Theorems on the uniqueness and thermal properties of stationary, nonsingular, quasifree states on space-times with a bifurcate Killing horizon*, Phys. Rep. **207**, 49 (1991).

[32] W. G. Unruh, *Experimental black hole evaporation?*, Phys. Rev. Lett. **46**, 1351 (1981).

[33] C. Barceló, S. Liberati, and M. Visser, *Analogue gravity*, Living Rev. Relativ. **29**, 1 (2026), arXiv:gr-qc/0505065.

[34] A. Coutant, R. Parentani, and S. Finazzi, *Black hole radiation with short distance dispersion, an analytical S-matrix approach*, Phys. Rev. D **85**, 024021 (2012), arXiv:1108.1821.